Neutron Radiation From IEC Fusion Reactor

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- Assume isotropic emission:

$$\phi(r) = \frac{Y}{4\pi r^2}$$

- Where Y is the neutron emission rate in neutrons/s and r is distance from the source.
- Assume worst conditions:

$$Y = 1{,}000{,}000\,\text{neutrons/s}, \qquad r = 1\,\text{m}$$

- My fusor is far less capable, and I will be farther than 1 m away.
- The neutron flux is:

$$\phi = \frac{1{,}000{,}000}{4\pi}\frac{\text{n}}{\text{s}\cdot\text{m}^2}$$

- Need a conversion factor from neutrons to Sieverts. Sieverts measure health risk from ionizing radiation while accounting for radiation type and tissue weighting.
- Use Table A.41 from ICRP Publication 116, page 199.
- Source: `https://journals.sagepub.com/doi/10.1016/S0146-6453%2896%2990010-X`

CONVERSION COEFFICIENTS FOR USE IN RADIOLOGICAL PROTECTION 199

Table A.41. Effective dose per unit neutron fluence, E/Φ, in units of pSv cm^2 for monoenergetic neutrons incident in various geometries on an adult anthropomorphic computational model. These reference values are presented graphically in Figs 22 and A.35 (Annex 1)

Energy (MeV)	AP	PA	RLAT	LLAT	ROT	ISO
1.0×10^{-9}	5.24	3.52	1.36	1.68	2.99	2.40
1.0×10^{-8}	6.55	4.39	1.70	2.04	3.72	2.89
2.5×10^{-8}	7.60	5.16	1.99	2.31	4.40	3.30
1.0×10^{-7}	9.95	6.77	2.58	2.86	5.75	4.13
2.0×10^{-7}	11.2	7.63	2.92	3.21	6.43	4.59
5.0×10^{-7}	12.8	8.76	3.35	3.72	7.27	5.20
1.0×10^{-6}	13.8	9.55	3.67	4.12	7.84	5.63
2.0×10^{-6}	14.5	10.2	3.89	4.39	8.31	5.96
5.0×10^{-6}	15.0	10.7	4.08	4.66	8.72	6.28
1.0×10^{-5}	15.1	11.0	4.16	4.80	8.90	6.44
2.0×10^{-5}	15.1	11.1	4.20	4.89	8.92	6.51
5.0×10^{-5}	14.8	11.1	4.19	4.95	8.82	6.51
1.0×10^{-4}	14.6	11.0	4.15	4.95	8.69	6.45
2.0×10^{-4}	14.4	10.9	4.10	4.92	8.56	6.32
5.0×10^{-4}	14.2	10.7	4.03	4.86	8.40	6.14
1.0×10^{-3}	14.2	10.7	4.00	4.84	8.34	6.04
2.0×10^{-3}	14.4	10.8	4.00	4.87	8.39	6.05
5.0×10^{-3}	15.7	11.6	4.29	5.25	9.06	6.52
1.0×10^{-2}	18.3	13.5	5.02	6.14	10.6	7.70
2.0×10^{-2}	23.8	17.3	6.48	7.95	13.8	10.2
3.0×10^{-2}	29.0	21.0	7.93	9.74	16.9	12.7
5.0×10^{-2}	38.5	27.6	10.6	13.1	22.7	17.3
7.0×10^{-2}	47.2	33.5	13.1	16.1	27.8	21.5
1.0×10^{-1}	59.8	41.3	16.4	20.1	34.8	27.2
1.5×10^{-1}	80.2	52.2	21.2	25.5	45.4	35.2
2.0×10^{-1}	99.0	61.5	25.6	30.3	54.8	42.4
3.0×10^{-1}	133	77.1	33.4	38.6	71.6	54.7
5.0×10^{-1}	188	103	46.8	53.2	99.4	75.0
7.0×10^{-1}	231	124	58.3	66.6	123	92.8
9.0×10^{-1}	267	144	69.1	79.6	144	108
1.0×10^{0}	282	154	74.5	86.0	154	116
1.2×10^{0}	310	175	85.8	99.8	173	130
2.0×10^{0}	383	247	129	153	234	178
3.0×10^{0}	432	308	171	195	283	220
4.0×10^{0}	458	345	198	224	315	250
5.0×10^{0}	474	366	217	244	335	272
6.0×10^{0}	483	380	232	261	348	282
7.0×10^{0}	490	391	244	274	358	290
8.0×10^{0}	494	399	253	285	366	297
9.0×10^{0}	497	406	261	294	373	303
1.0×10^{1}	499	412	268	302	378	309
1.2×10^{1}	499	422	278	315	385	322
1.4×10^{1}	496	429	286	324	390	333
1.5×10^{1}	494	431	290	328	391	338
1.6×10^{1}	491	433	293	331	393	342
1.8×10^{1}	486	435	299	335	394	345
2.0×10^{1}	480	436	305	338	395	343
3.0×10^{1}	458	437	324	na[a]	395	na[a]
5.0×10^{1}	437	444	358	na	404	na
7.5×10^{1}	429	459	397	na	422	na
1.0×10^{2}	429	477	433	na	443	na
1.3×10^{2}	432	495	467	na	465	na
1.5×10^{2}	438	514	501	na	489	na
1.8×10^{2}	445	535	542	na	517	na

Effective dose per unit neutron fluence, E/Φ, from ICRP Publication 116 Table A.41.

- The AP column means anterior-to-posterior, which is what you want if a person is facing the radiation source.
- From the table:

$$2.0\,\text{MeV} : 383\,\text{pSv}\,\text{cm}^2/\text{neutron}$$

$$3.0\,\text{MeV} : 432\,\text{pSv}\,\text{cm}^2/\text{neutron}$$

- D-D fusor neutrons are 2.45 MeV, so interpolate:

$$h \approx 383 + 0.45(432 - 383)$$

$$h \approx 405\,\text{pSv}\,\text{cm}^2/\text{neutron}$$

- Convert units:

$$405\,\text{pSv}\,\text{cm}^2/\text{neutron} = 405 \times 10^{-12}\,\text{Sv} \times 10^{-4}\,\text{m}^2/\text{neutron}$$

$$h \approx 4.05 \times 10^{-14}\,\text{Sv}\,\text{m}^2/\text{neutron}$$

- So we use the scaling factor:

$$h \approx 4 \times 10^{-14}\,\text{Sv}\,\text{m}^2/\text{neutron}$$

- Plug back in:

$$\dot{E} \approx \frac{1{,}000{,}000}{4\pi} \frac{\text{n}}{\text{s} \cdot \text{m}^2} \cdot 4 \times 10^{-14} \frac{\text{Sv}\,\text{m}^2}{\text{n}}$$

- Simplify:

$$\dot{E} \approx \frac{1{,}000{,}000}{4\pi} \cdot 4 \times 10^{-14} \frac{\text{Sv}}{\text{s}}$$

$$\dot{E} \approx 3.18 \times 10^{-9}\,\text{Sv/s}$$

$$\dot{E} \approx 3.18\,\text{nSv/s}$$

- Scale to hours:

$$\dot{E} \approx 3.18\,\text{nSv/s} \cdot 3600\,\text{s/hr}$$

$$\dot{E} \approx 11.5\,\mu\text{Sv/hr}$$

- For context, the NRC public-dose limit for licensed operations is 1 mSv/yr and the unrestricted-area external dose-rate limit is 20 μSv/hr.
- NRC source: `https://www.law.cornell.edu/cfr/text/10/20.1301`
- Therefore, even a relatively high-yield homemade fusor at 10^6 neutrons/s gives a neutron dose rate below the NRC unrestricted-area hourly dose-rate limit at 1 m away.
- In fact, natural background radiation is about 0.3 μSv/hr, per the CDC.
- So for every hour you operate a high-powered fusor from 1 m away, you are exposed to about 38 hours of natural background radiation. Nothing.
- This also scales linearly with neutrons/s. So if we assume a more reasonable estimate of 50,000 neutrons/s for my fusor instead of 1,000,000 neutrons/s, exposure drops by a factor of 20:

$$\dot{E} \approx \frac{11.5}{20}\,\mu\text{Sv/hr}$$

$$\dot{E} \approx 0.575\,\mu\text{Sv/hr}$$

  when standing 1 m away.

- Distance scales as $1/r^2$, so going from 1 m to 3 m reduces exposure by a factor of:

$$3^2 = 9$$

- Therefore:

$$\dot{E}_{3\text{m}} = \frac{0.575}{9}\,\mu\text{Sv/hr}$$

$$\dot{E}_{3\text{m}} \approx 0.064\,\mu\text{Sv/hr}$$

  when standing 3 m away at 50,000 neutrons/s.

- This is the most reasonable estimate for actual operational neutron exposure from my fusor.
- Conclusion: Neutron radiation from my fusor is negligible past 1 m away from the fusor. Just don't hug it at peak operation.